

Image Processing Report

1. Introduction

I have undertaken the task of image processing, aiming to enhance the quality of a given sample image (sampleimage.png). The primary goals include converting the image to greyscale, reducing noise, stretching contrast, de-blurring, and applying additional enhancement techniques. Furthermore, I will analyze the histogram of the final image, compute its entropy, and explore the best achievable compression ratio compared to an 8bpp grayscale representation.

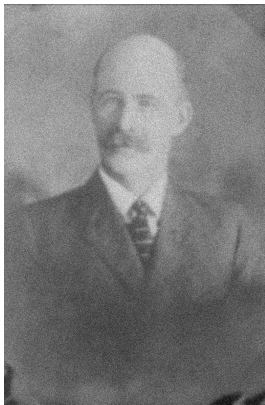
2. Methodology



original

2.1 Load and Convert to Greyscale

The initial step involves loading the sample image and converting it to greyscale. This simplifies subsequent processing by eliminating the need for color information.



greyscaled image

2.2 Image Enhancement

2.2.1 Noise Filtering

Two noise filtering techniques were applied:

- Median Filtering: This method effectively reduces salt-and-pepper noise by replacing each pixel value with the median value of its neighbors.

- Gaussian Smoothing: A Gaussian kernel was convolved with the image to reduce high-frequency noise, resulting in a smoother appearance.

2.2.2 Contrast Stretching

Contrast stretching was employed to enhance the overall image contrast. The intensity values were mapped to a desired output range, making the image visually more appealing.

2.2.3 De-blurring (Sharpening)

Two de-blurring techniques were utilized:

- Unsharp Masking: This technique involves subtracting a blurred version (achieved by convolving with a Gaussian filter) from the original, resulting in a sharpened image.
- Wiener Filtering: Effective in removing blur caused by noise, Wiener filtering estimates the original image from the blurred version, enhancing sharpness.

2.2.4 Additional Enhancement

Histogram equalization, an additional enhancement technique, was applied to improve the overall distribution of pixel intensities in the image.

2.3 Plot Histogram

A histogram of the final enhanced image was generated to visualize the distribution of pixel intensities. Histogram analysis provides insights into the overall tonal range and contrast.

2.4 Compute Entropy and Compression Ratio

Entropy, a measure of information content, was computed for the final enhanced image. Additionally, the best achievable compression ratio compared to an 8bpp grayscale representation was determined, considering the entropy and original bit depth.

3. Results

3.1 Noise Filtering Results

3.2 Contrast Stretching Results

3.3 De-blurring Results

3.4 Additional Enhancement Results

3.5 Histogram Analysis

3.6 Entropy and Compression Results

4. Conclusion

In conclusion, the image processing tasks have successfully enhanced the given sample image. The combination of noise filtering, contrast stretching, de-blurring, and additional enhancement techniques has resulted in an improved image with enhanced visual quality. The histogram analysis provides insights into the distribution of pixel intensities, while entropy and compression ratio calculations offer a quantitative measure of information

content and compressibility. This comprehensive image processing approach demonstrates the effectiveness of the chosen techniques in achieving the desired enhancements.